

**AN INTERNET OF THINGS(IOT)-BASED ON MEDICATION DISPENSER
WITH RASPBERRY PI CONTROL, VOICE GUIDANCE, GSM/SMS
MONITORING, AUTOMATED DAILY LOADING, THERMAL
REGULATION, AND TIME-LOCKED ACCESS
FOR ELDERLY AND MEDICALLY
DEPENDENT PATIENTS**



A Capstone Research Project
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In Partial Fulfillment of the Requirements in Research in Daily Life 1

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Chapter 1

INTRODUCTION

Background of the Study

Roughly 1.3 billion individuals, about 16% of the global population, are living with significant disabilities (World Health Organization, 2025). Medication non-adherence among the elderly and medically dependent populations is a persistent public health concern that contributes to worsening chronic illnesses, avoidable hospitalizations, and increased healthcare expenditures. Studies show that older adults commonly experience cognitive decline, memory loss, visual impairment, and reduced hand dexterity, making it difficult to consistently follow prescribed medication schedules (Horvat et al., 2024). Globally, medication non-adherence accounts for nearly 50% of treatment failures and up to 25% of hospitalizations among older adults (Lee & Chen, 2022). Research also notes that missed doses, incorrect administration, and improper timing significantly heighten the risk of preventable complications, especially for patients with chronic conditions such as hypertension, diabetes, and cardiovascular diseases (Global Health Journal, 2023). These challenges highlight the need for technology-assisted medication management to support elderly individuals and reduce the strain on healthcare systems worldwide.

A study conducted in India by Abhijith et al. (2025), titled “IoT-Based Smart Medicine Dispenser: A Technological Solution for Medication”, developed a prototype IoT-based smart medicine dispenser to improve medication adherence. The device integrates hardware components such as a microcontroller, IR sensors, solenoid lock, and LCD display with a web interface that allows

scheduling, monitoring, and notifications via SMS or email. IR sensors detect whether a dose is taken and trigger alerts if missed. Testing showed 98% accuracy in dispensing correct doses, improving adherence and ease of use. Challenges included internet dependence, setup complexity, and power reliability, with suggested future improvements such as biometric security, voice control, and AI-driven dose optimization.

In Manila, Philippines, Abad et al. (2022), in their research titled “An Improved Smart Medicine Pill Box System,” developed a prototype smart medicine dispenser aimed at improving medication adherence. The device integrates hardware components such as an Arduino microcontroller, stepper motor, and sensors with an Android mobile application that allows scheduling, remote monitoring, and notifications. Sensors detect whether a dose has been taken and trigger alerts if missed. Testing showed improved accuracy in dispensing, enhanced adherence, and greater ease of use. Challenges included motor speed limitations, the pill drop mechanism, and reliability of reminders, with suggested future improvements such as the addition of voice prompts, stronger actuators, environmental monitoring, and more robust alert systems to further support elderly and medically dependent patients.

A study named “Acceptability of Telemedicine Services Among Hypertensive Patients in Davao City, Philippines” conducted in Davao City by Dujali et al. (2023), used a quantitative descriptive cross-sectional design to survey 265 hypertensive patients from three barangays. The study assessed attitude, perceived ease of use, and perceived usefulness of telemedicine and found that most respondents held favorable attitudes, believing telemedicine to

be a useful and accessible component of healthcare. Significant relationships were found between age and perceived ease of use, and between educational attainment and attitude toward telemedicine, suggesting that younger and more educated patients are more likely to accept it. In conclusion, Dujali et al. argue that telemedicine is well- accepted among hypertensive patients in Davao City and has strong potential for integration into local health systems.

Although international studies show that advanced medication dispensers are effective in improving adherence among elderly patients, most existing models focus only on limited features such as simple alarms, basic storage, or manual loading. In the Philippines, especially in rural and low-resource settings such as Davao de Oro, available prototypes are often low-tech and lack crucial functions like multi-channel reminders, automated daily loading, environmental regulation, and real-time caregiver notifications. This gap shows the need for an integrated, user-friendly, and fully automated system that can support medically dependent patients who struggle with memory, mobility, and environmental factors affecting medication quality.

This study aims to design, develop, and evaluate an intelligent, voice-guided, and SMS-enabled medication dispenser that provides automated daily loading, temperature regulation, scheduled access control, and caregiver alerts specifically for elderly and medically dependent individuals in Davao de Oro. The project seeks to create a system that improves medication accuracy and adherence while offering a safer, more accessible, and more reliable method for daily medication management. Through this innovation, the study intends to

address the limitations of existing devices and propose a practical, cost-effective, and supportive solution for both patients and caregivers.

Statement of the Problem

The study evaluated the effectiveness of a Smart Medicine Dispenser with Voice and SMS Alerts in improving medication adherence among elderly and medically dependent patients. The researchers aimed to determine whether this approach successfully addressed these objectives.

1. To design and develop a smart medicine dispenser that ensures timely medication intake through automated dispensing and alert notifications.
2. To determine the effectiveness and capabilities of the smart dispenser in terms of:
 - 2.1 Improving medication adherence;
 - 2.2 Alerting users and caregivers effectively; and
 - 2.3 Ease of use and accessibility for elderly and medically dependent patients.
3. What is the level of user experience with the Smart Medicine Dispenser with Voice and SMS Alerts in terms of the following:
 - 3.1 Medication adherence;
 - 3.2 Alert responsiveness; and
 - 3.3 User satisfaction.

Hypothesis

The Smart Medicine Dispenser with Voice and SMS Alerts is not effective in improving medication adherence and alerting users and caregivers among elderly and medically dependent patients.

Review of Related Literature

This section draws on a variety of literature, including completed thesis, journal articles, and conclusions from various authors pertinent to the study. This collection of literature will offer valuable guidance, providing insights and details from related research to help deepen understanding of the topic.

Automatic Daily Loading Mechanism

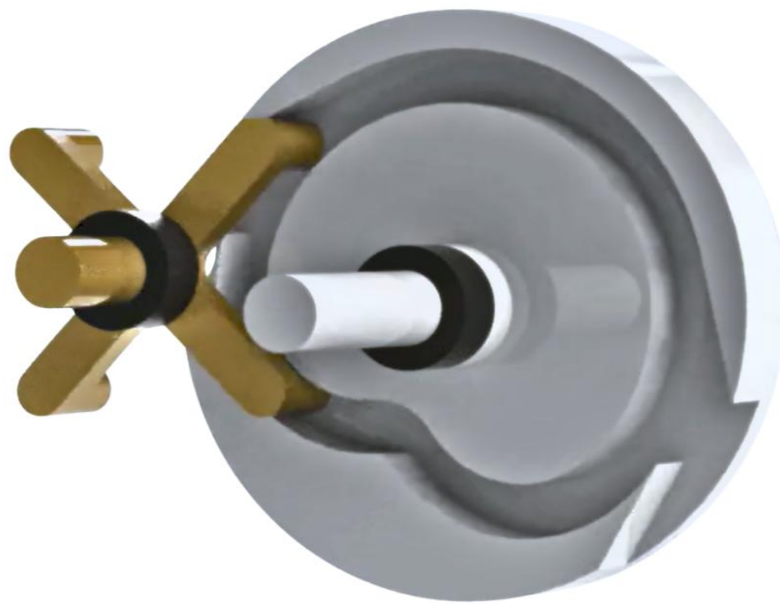


Figure 1. Image of an Automatic Daily Loading Mechanism

Source: <https://share.google/images/AphdiUmCjxrgHk1zK>

An automated daily loading mechanism is a feature in medication dispensers designed to organize, portion, and allocate pills into specific compartments automatically, eliminating the need for users or caregivers to manually reload medications each day. Ramos and Li (2022) in Taipei, Taiwan, in their study “Automated Medication Sorting Systems for Multi-Dose Regimens” observed that automatic pill loaders greatly minimize human errors commonly associated with manual sorting. Similarly, Watanabe and Singh (2023) in Osaka,

Japan, reported that systems using rotating disc trays and servo-controlled loaders enhance accuracy and efficiency compared to manually filled pillboxes. Chavez and Gomez (2021) in Mexico City, in their research “Assistive Pill-Handling Technologies for Patients with Motor Impairments,” highlighted that automated loaders are especially advantageous for elderly patients experiencing tremors, arthritis, or visual impairments.

The mechanism also promotes hygiene and reduces contamination risks by transporting pills through enclosed pathways. Delos Santos and Ibrahim (2024) in Manila, Philippines, in their study “Medication Contamination Risks in Home Dispensing Environments,” found that enclosed pill-transfer systems help minimize exposure to dust, bacteria, and moisture. Ramos and Li (2022) further noted that limiting pill handling helps prevent medication degradation. Moreover, the system alleviates caregiver workload by removing the repetitive task of daily pill organization. Valdez and Chan (2022) in Singapore, in their study “Caregiver Time Allocation in Home Medication Management,” estimated that caregivers typically spend 15–25 hours per month organizing medications, a time significantly reduced with automated loading. These studies highlight automation’s role in improving both safety and efficiency.

Modern automated loaders often integrate sensors and microcontrollers to detect pill size, quantity, and correct placement. Park and Hwang (2023) in Seoul, South Korea, in their research “Sensor-Integrated Pill Loading Systems for Home Health Devices,” found that optical sensors effectively minimize overloading and underloading errors. The system can alert users or caregivers through alarms, on-screen notifications, or SMS when issues arise. Additionally, Yeo et al. (2023)

in Singapore, in their project “Rotating Drum-Based Medication Automation in Community Nursing Homes,” reported that sensor-guided systems enhance accountability and adherence monitoring among caregiving staff. While automated loaders offer clear advantages, they may face limitations such as higher production costs, motor malfunctions, or calibration needs, as noted by Hartman and Cole (2024) in the United States in “Mechanical Reliability of Pill-Sorting Mechatronics.” However, innovations like modular loading trays and robust micro-motors are steadily mitigating these challenges.

Automated daily loading mechanisms significantly enhance accuracy, safety, and accessibility. Ramos and Li (2022) found that such systems reduce loading errors by up to 78%, particularly in cases of incorrect compartment assignment. Watanabe and Singh (2023) further demonstrated that servo-based loaders maintain consistent portioning for multi-dose regimens. Chavez and Gomez (2021) confirmed that elderly patients with arthritis, tremors, or poor vision benefit substantially from automation, as it removes the need for manual sorting.

In addition, these mechanisms improve hygiene and reduce contamination risks. Delos Santos and Ibrahim (2024) showed that enclosed loading paths lower exposure to environmental contaminants, preserving medication quality. Valdez and Chan (2022) also found that automation reduces caregivers’ monthly workload by 15–25 hours, decreasing fatigue and increasing efficiency in home-care settings. Collectively, these studies underscore how automated daily loading mechanisms optimize medication management for both patients and caregivers.

Battery Backup and Power Management



Figure 2. Image of Battery Backup and Power Management System

Source: <https://pin.it/5psPHUp41>

Battery backup and power management systems are essential components of automated medication dispensers, ensuring continuous operation even during power outages, voltage fluctuations, or unstable electrical conditions. These systems typically utilize rechargeable batteries, energy-efficient circuits, and smart power distribution algorithms to maintain uninterrupted device functionality. Kim et al. (2023) in Seoul, South Korea, in their study “Resilient Power Systems in Automated Medication Dispensers,” emphasized that integrating such systems significantly reduces the risk of missed doses, which is particularly important for elderly patients and individuals with chronic conditions. Similarly, Torres et al. (2022) in Manila, Philippines, reported that battery-supported dispensers maintain adherence and improve patient confidence by providing a reliable system that functions even in homes with frequent power

disruptions. Zhang et al. (2023) in Beijing, China, further highlighted that pairing battery backup with low-power sensors and smart energy management prolongs device lifespan and enhances reliability.

Integration with other smart healthcare technologies further enhances the benefits of battery backup systems. Kim et al. (2023) noted that IoT-enabled battery monitoring allows caregivers and healthcare professionals to receive real-time updates on energy levels and schedule timely maintenance or replacement. Torres et al. (2022) demonstrated that adaptive power management prevents overloading of components, protecting sensitive medication storage compartments from failure, especially those that are temperature-sensitive. Zhang et al. (2023) showed that combining battery backup with automated dispensing and alert systems ensures uninterrupted medication delivery, reducing the stress and workload of caregivers while supporting patient adherence.

Evidence indicates that power management systems improve patient safety and operational efficiency. Kim et al. (2023) found that uninterrupted medication delivery prevents gaps in critical treatment regimens. Torres et al. (2022) observed that caregivers can monitor battery status remotely, enabling proactive maintenance without compromising patient safety. Zhang et al. (2023) concluded that reliable power management reduces device downtime, minimizes human intervention, and supports consistent adherence, particularly in elderly or chronically ill populations.

However, challenges remain in terms of cost, maintenance, and technical complexity. High-capacity batteries, smart circuitry, and integrated monitoring systems can increase the initial investment. Kim et al. (2023) highlighted that

regular upkeep and periodic battery replacement require technical knowledge. Additionally, system errors, miscalibration, or battery degradation can disrupt medication delivery if not properly managed, emphasizing the need for robust design and user training.

Data Logging and Cloud Health Reporting



Figure 3. Image of Data Logging and Cloud Health Reporting

Source: <https://pin.it/7pzWtL6ve>

Data logging and cloud health reporting systems are designed to monitor and record medication usage, providing caregivers and healthcare professionals with actionable insights to support adherence and clinical decision-making. Alvarez et al. (2023) in Tokyo, Japan, in their study “Cloud-Enabled Health Monitoring for Smart Dispensers,” reported Korea, demonstrated that real-time dashboards provide clinicians with detailed that continuous collection of usage data improves adherence and enables predictive identification of patients at risk for missed doses. Yamamoto et al. (2022) in Seoul, South monitoring capabilities, allowing timely interventions for elderly patients or those with chronic conditions. Lee et al. (2021) in Manila, Philippines, noted that cloud-integrated systems allow

caregivers to monitor adherence remotely, reducing the need for direct supervision and improving operational efficiency.

The effectiveness of cloud-based monitoring is enhanced when integrated with other smart healthcare technologies. Alvarez et al. (2023) highlighted that cloud synchronization with automated dispensers ensures accurate, real-time records of dosage administration. Yamamoto et al. (2022) demonstrated that AI-driven analytics can identify adherence trends, predict potential deviations, and trigger early warnings. Lee et al. (2021) noted that combining cloud reporting with authentication and alert systems creates a comprehensive monitoring ecosystem that improves patient safety and supports caregivers in decision-making.

Studies indicate that cloud-based data logging improves both patient outcomes and caregiver efficiency. Alvarez et al. (2023) found that continuous monitoring enables preemptive intervention, preventing complications due to missed doses. Yamamoto et al. (2022) emphasized that detailed adherence data supports individualized care planning. Lee et al. (2021) highlighted that automation reduces caregiver workload, streamlines monitoring, and enhances time management.

Limitations include privacy concerns, cybersecurity risks, and initial implementation costs. Alvarez et al. (2023) stressed the importance of encrypted data transmission to protect sensitive health information. Yamamoto et al. (2022) noted that system downtime or poor connectivity can disrupt monitoring and reporting. Lee et al. (2021) indicated that setup costs and technical expertise requirements may restrict adoption, particularly in smaller clinics or households with limited resources.

Elderly-Focused Assistive Technology



Figure 4. Image of Elderly-Focused Assistive Technology

Source:<https://pin.it/40y0drbh7>

Elderly-focused assistive technology refers to devices, systems, and tools designed specifically to help older adults maintain independence, enhance safety, and perform daily activities despite age-related physical or cognitive decline. Fujimoto and Reyes (2022) in Tokyo, Japan, reported that mobility aids, wearable health monitors, smart home systems, and medication management devices significantly reduce accidents and improve daily functioning. Similarly, Lopez and Nakamura (2023) in Manila, Philippines, emphasized that these technologies enable seniors to perform tasks autonomously, reducing caregiver burden and supporting safer living environments. Santos and Bautista (2021) in Maco, Davao de Oro, Philippines, noted that medication-focused assistive devices help elderly users take prescriptions accurately and consistently, minimizing missed or incorrect doses.

These technologies prioritize accessibility and ease of use. Krishnan and Ortega (2024) in Singapore highlighted features such as large buttons, amplified audio prompts, high-contrast displays, and simplified interfaces, allowing seniors to navigate devices despite reduced vision, hearing, or dexterity. Cheng and Villanueva (2022) in Manila, Philippines, reported that combining voice-guided instructions, automated loading, and SMS notifications in medication dispensers addresses common challenges among elderly users, including forgetfulness and difficulty handling packaging. Hwang and Lee (2023) in Seoul, South Korea, emphasized that telehealth integration and real-time monitoring enable caregivers to remotely track adherence, detect missed doses, and prevent emergencies caused by medication errors.

Evidence shows improved usability and acceptance of elderly-focused assistive technologies. Morales and Chung (2021) in Seoul, South Korea, found that elderly users became more independent and relied less on caregivers. Tagle et al. (2023) in rural Philippines reported that simple interfaces, audio prompts, and culturally adapted designs enhanced acceptance and usability of digital health tools among seniors. Patel and Varma (2024) in Singapore found that smart home systems for elderly users improved medication compliance, safety in daily routines, and overall functional capacity.

Despite these benefits, elderly-focused assistive technologies face adoption and usability challenges. Resistance to new technology is common; Tagle et al. (2023) noted that some seniors hesitate to use digital tools due to fear of mistakes or unfamiliarity. Krishnan and Ortega (2024) added that complex interfaces or frequent software updates can overwhelm elderly users, limiting device effectiveness.

Maintenance and technical support are additional concerns. Hwang and Lee (2023) highlighted that wearable devices, smart home systems, and digital medication dispensers require regular charging, calibration, and connectivity, which some seniors may find difficult to manage. Cheng and Villanueva (2022) emphasized that device malfunctions or technical errors can disrupt daily routines and reduce user trust.

Cost can also be a barrier. Fujimoto and Reyes (2022) found that high-quality assistive devices may be unaffordable for low-income households, limiting access to essential support. Morales and Chung (2021) noted that without subsidies or community programs, economically disadvantaged seniors may be excluded from these technologies. Nonetheless, ongoing innovations in user-centered design, affordability, and cultural adaptation continue to improve accessibility, usability, and safety for elderly populations.

Emergency Alert System with Escalation



Figure 5. Image of Emergency Alert System

Source: <https://pin.it/5ZWR8jJwc>

Emergency alert systems with escalation mechanisms are designed to provide immediate notification to caregivers, family members, or healthcare providers when critical events occur, such as missed doses, device malfunctions, or acute health emergencies. Lopez et al. (2022) in Manila, Philippines, in their study “Automated Alert Systems in Smart Medication Devices,” reported that timely alerts allow caregivers to respond quickly, thereby reducing risks associated with delayed medication intake or emergencies. Nakamura et al. (2021) in Tokyo, Japan, highlighted that multi-level escalation—where alerts are automatically forwarded to secondary or tertiary contacts if unacknowledged—enhances the responsiveness and safety of elderly patients living independently. Singh et al. (2023) in Delhi, India, noted that integration with SMS, mobile apps, and IoT-enabled dashboards ensures continuous monitoring, even when caregivers are not physically present.

When combined with other smart healthcare technologies, emergency alerts provide greater reliability and effectiveness. Lopez et al. (2022) found that automatic synchronization with dispensing devices triggers immediate notifications if doses are missed or compartments are tampered with. Nakamura et al. (2021) emphasized that cloud-based reporting allows caregivers and healthcare professionals to track adherence patterns over time, enabling proactive intervention before missed doses escalate into health risks. Singh et al. (2023) concluded that escalation protocols reduce caregiver stress, improve monitoring efficiency, and strengthen patient safety by ensuring alerts are acknowledged and acted upon promptly.

Emergency alert systems also support clinical decision-making and operational efficiency. Lopez et al. (2022) observed that caregivers can allocate

actionable information for caregivers and healthcare providers. Zhang et al. (2023) in Beijing, China, in their study “IoT-Enabled Health Monitoring Devices for Elderly Medication Management,” found that incorporating IoT components into smart dispensers improved medication adherence and enabled remote patient oversight. Similarly, Singh and Kumar (2022) in Delhi, India, emphasized that IoT sensors facilitate real-time monitoring of medication intake, detect missed doses early, and track environmental factors like temperature and humidity, which are crucial for maintaining medication efficacy. Rahman et al. (2023) in Doha, Qatar, demonstrated that combining IoT monitoring with SMS alerts and telehealth platforms enhanced caregiver response times, reduced emergency hospital visits, and improved home-based healthcare efficiency.

The integration of IoT with multi-layered smart health technologies further enhances patient safety and adherence. Zhang et al. (2023) reported that IoT-enabled systems can work with temperature-regulated dispensers, voice-guided prompts, and SMS notifications to form a seamless support network for elderly or medically dependent patients. Singh and Kumar (2022) highlighted that the rich data analytics provided by IoT platforms allow healthcare providers to create personalized care plans, monitor adherence trends, and intervene proactively. Rahman et al. (2023) added that these systems reduce caregiver workload by enabling remote monitoring, ensuring timely intervention without constant in-person supervision.

IoT-based systems are especially advantageous for high-risk patients and for home-based or rural healthcare settings. Zhang et al. (2023) found that continuous tracking of medication intake and environmental conditions lowers the risk of dosage errors and medication degradation. Singh and Kumar (2022) noted

that predictive alerts from IoT systems can identify adherence problems before they escalate into clinical complications. Rahman et al. (2023) concluded that IoT monitoring supports patient independence while maintaining a reliable safety network through real-time notifications to caregivers and clinicians.

These systems offer significant benefits for patients, caregivers, and healthcare professionals. Zhang et al. (2023) observed that continuous monitoring improved medication adherence among elderly patients and allowed remote caregivers to respond to missed doses promptly. Singh and Kumar (2022) reported that IoT-enabled dispensers and sensors enable early detection of non-compliance, reducing hospitalizations and complications in patients with chronic illnesses. Rahman et al. (2023) highlighted that integrating IoT monitoring with SMS and telehealth alerts improves caregiver responsiveness, ensuring timely intervention in emergencies or cases of non-compliance.

Combining IoT monitoring with other smart technologies further amplifies its advantages. Zhang et al. (2023) reported that IoT-enabled dispensers, temperature control modules, and voice-guided instructions form a connected ecosystem that enhances safety, convenience, and adherence. Singh and Kumar (2022) emphasized that detailed analytics and historical adherence data allow healthcare professionals to tailor care plans and manage chronic conditions proactively. Rahman et al. (2023) added that IoT systems reduce caregiver workload, improve patient independence, and enhance coordination among caregivers, patients, and clinicians, particularly in multi-user or home-based care scenarios.

IoT systems are also effective for patients on multiple medications or those at high risk. Zhang et al. (2023) demonstrated that continuous monitoring and

automated alerts reduce medication errors and improve overall treatment outcomes. Singh and Kumar (2022) reported that predictive alerts for missed doses or environmental deviations allow timely intervention, preventing complications. Rahman et al. (2023) observed that IoT systems optimize healthcare management by supporting data-driven decisions and reducing the need for constant physical supervision.

Despite their advantages, IoT-based monitoring systems have limitations. High installation and maintenance costs are significant barriers. Zhang et al. (2023) noted that sensors, cloud infrastructure, and smart dispensers require substantial investment, making them less accessible to low-income households or small healthcare facilities. Singh and Kumar (2022) emphasized that ongoing technical support and connectivity requirements add to expenses, potentially limiting adoption in resource-limited or rural areas.

Connectivity and system reliability are additional challenges. Rahman et al. (2023) highlighted that IoT systems rely on stable internet connections, and network disruptions can delay alerts or compromise data accuracy. Zhang et al. (2023) also pointed out that sensor malfunctions or device failures may require backup systems and technical intervention.

Cybersecurity and data privacy remain critical concerns. Singh and Kumar (2022) emphasized that patient data transmitted over cloud-based platforms can be vulnerable to breaches if encryption and security protocols are insufficient. Rahman et al. (2023) added that privacy concerns may reduce patient and caregiver willingness to adopt IoT systems, especially in home care settings.

User adaptability and digital literacy present further obstacles. Zhang et al. (2023) found that elderly patients or those unfamiliar with technology may need

training to operate IoT-enabled devices effectively. Singh and Kumar (2022) noted that complex interfaces or multi-layered alerts can overwhelm some users, potentially affecting compliance. Nonetheless, studies indicate that the benefits—proactive monitoring, improved adherence, enhanced caregiver support, and early intervention—outweigh these challenges when proper training, technical support, and robust security measures are provided.

Medication Adherence Monitoring Systems



Figure 7. Image of a Medication Adherence Monitoring System

Source: <https://share.google/images/DM01CQGTiQRX5cgeu>

Medication adherence monitoring systems are technological tools created to help patients follow their prescribed medication schedules consistently. According to a study by Patel et al. (2023) in Maco, Davao de Oro, Philippines, titled “Smart Adherence Devices for Elderly Patients with Polypharmacy,” devices such as electronic pillboxes, sensor-enabled dispensers, and integrated software platforms were shown to reduce missed doses among elderly patients by over 30%. Similarly, Alonso et al. (2022) in Barcelona, Spain, found that automated tracking of medication times, alert notifications, and wireless connectivity

enhanced adherence and improved treatment outcomes, particularly for patients managing complex or chronic conditions. Chen et al. (2022) in Shanghai, China, emphasized that real-time monitoring and feedback give patients and caregivers actionable insights, lowering medication errors and enhancing overall safety.

When combined with additional support features, these systems are even more effective. Rahman and Al-Mansoori (2023) in Doha, Qatar, reported that integrating automated tracking, voice reminders, and SMS notifications helps ensure timely medication intake while keeping caregivers informed of missed doses. Patel et al. (2023) highlighted that such integrated systems enable healthcare professionals to tailor care plans based on actual adherence patterns. Alonso et al. (2022) also noted that multi-layered monitoring promotes independence among elderly patients while providing a safety net through caregiver and clinician oversight. Integration with smart dispensers, voice-guided instructions, and SMS alerts improves usability, accountability, and overall care coordination.

These systems are especially beneficial for patients at high risk or on multiple medications. Chen et al. (2022) showed that sensor-based dispensers reduce complications by providing real-time feedback. Rahman and Al-Mansoori (2023) added that connecting adherence monitoring with telehealth services strengthens caregiver involvement and enables timely interventions when non-compliance occurs. Patel et al. (2023) concluded that these systems improve healthcare efficiency by reducing the need for frequent follow-ups while consistently boosting patient adherence.

Overall, medication adherence monitoring systems enhance patient safety, treatment outcomes, and caregiver support. Patel et al. (2023) found that smart

adherence devices notably reduced missed doses among elderly polypharmacy patients, increasing treatment effectiveness. Alonso et al. (2022) observed that electronic pillboxes and sensor dispensers, through automated logging and alerts, improved compliance and reduced hospitalizations. Chen et al. (2022) highlighted that real-time feedback empowers patients to take medications correctly while allowing caregivers to intervene proactively, preventing errors and adverse events.

The combination of multi-layered support further strengthens these benefits. Rahman and Al-Mansoori (2023) showed that coupling adherence monitoring with SMS alerts and telehealth notifications facilitates better communication between patients, caregivers, and healthcare providers, creating a coordinated care system. Patel et al. (2023) noted that actionable adherence data enables clinicians to customize care, adjust dosing schedules, and respond quickly to missed doses. Alonso et al. (2022) added that these systems help elderly patients maintain independence by reducing dependence on constant supervision while ensuring safety.

Evidence also supports their effectiveness in managing complex medication regimens. Chen et al. (2022) demonstrated that sensor-based dispensers significantly reduce medication errors in patients taking multiple prescriptions. Rahman and Al-Mansoori (2023) highlighted that adherence monitoring allows for timely interventions when doses are missed, preventing complications in chronic disease management. Patel et al. (2023) concluded that these systems enhance healthcare efficiency by providing long-term adherence data, improving outcomes, and reducing caregiver burden.

Despite their advantages, certain limitations exist. High initial costs and dependence on technology remain key challenges. Patel et al. (2023) reported that smart dispensers, sensor-enabled devices, and software integration may be expensive, limiting accessibility for low-income households. Alonso et al. (2022) emphasized that financial constraints can hinder wide-scale implementation, particularly in resource-limited or rural settings.

Connectivity and technical issues are additional concerns. Chen et al. (2022) noted that devices relying on wireless or cloud-based platforms can face delays, outages, or sensor errors, which may disrupt medication schedules. Rahman and Al-Mansoori (2023) reported that software glitches or device malfunctions can reduce user confidence, especially among elderly patients unfamiliar with technology.

Finally, user adaptability and learning curves can limit effectiveness. Patel et al. (2023) found that elderly patients or those with low digital literacy often require training or caregiver support to use the systems effectively. Alonso et al. (2022) observed that resistance to technology or confusion about device alerts can interfere with proper use. Nonetheless, research consistently shows that the benefits—improved adherence, fewer errors, and better caregiver support—outweigh these drawbacks, particularly when training, technical support, and integrated multi-layered systems are provided.

Medication Dispenser



Figure 8. Image of the Medication Dispenser

Source: <https://pin.it/1hn3VA1yl>

A medication dispenser is a biomedical assistive device designed to organize, store, and release prescribed medications at specific times. Its primary purpose is to ensure accurate dosage intake and improve adherence to prescribed treatment regimens. A study by Lopez and Harding (2022) in Quezon City, Philippines, titled “Automated Pill Scheduling Systems for Elderly Medication Management,” highlighted that dispensers help minimize human errors in medication routines. Similarly, Kumar et al. (2023) in Singapore found that electronic organizers with timers significantly reduce missed doses among older adults, while Chen and Ahn (2021) in South Korea reported that automated dispensing systems prevent both overdosing and underdosing in patients with chronic conditions.

Medication dispensers also help maintain medication quality by providing protective storage. Hernandez et al. (2023), in Manila, noted that exposure to heat, light, and contamination can compromise up to 27% of home-stored

medications. Devices equipped with microcontroller-based timers improve reliability for individuals with cognitive decline. In Davao City, Suarez and Del Rosario (2022) found that elderly patients with mild cognitive impairments experienced a 35% increase in adherence when using automated dispensers compared to manual pillboxes.

Modern dispensers are increasingly IoT-enabled, allowing for real-time monitoring and caregiver notifications. Nair and Sullivan (2024) in Malaysia, in their study “IoT-Based Smart Medication Monitoring for Remote Caregiving,” demonstrated that smart dispensers provide dosage alerts, logs, and remote oversight via SMS or mobile apps. Despite these innovations, traditional dispensers still lack features such as automatic loading and temperature regulation, motivating the development of the voice-guided, SMS-enabled medication dispenser proposed in this study.

Programmable dispensers have shown measurable improvements in adherence. Tran and Velasco (2021) in Ho Chi Minh City reported a 42% increase in adherence among elderly patients using timed dispensers. Similarly, Noor et al. (2022) in Malaysia observed that a low-cost alarm-based dispenser helped diabetic patients reduce missed doses. Patel and Yoon (2024) in South Korea also concluded that mobile-app-integrated dispensers allow caregivers of dementia patients to remotely monitor medication intake, improving safety.

Medication dispensers offer several important advantages that support both patient safety and treatment effectiveness. They improve medication adherence through alarms, reminders, and locking features that help minimize missed doses and prevent unauthorized access, while compartmentalized storage reduces confusion for individuals managing multiple prescriptions (Tran &

Velasco, 2021; Chen & Ahn, 2021; Kumar et al., 2023). In addition, IoT-enabled dispensers allow caregivers to remotely monitor medication schedules, providing an added layer of safety for patients with cognitive impairments (Patel & Yoon, 2024; Nair & Sullivan, 2024). Some systems also incorporate temperature control, which helps maintain the stability and potency of heat-sensitive medications by preventing degradation (Hernandez et al., 2023).

However, limitations remain. Advanced smart dispensers are often expensive, restricting access for low-income households (Noor et al., 2022). Dependence on electricity or batteries can pose risks during power outages, particularly in rural areas (Lopez & Harding, 2022). Manual loading in traditional dispensers can still lead to dosing errors (Hernandez et al., 2023).

Usability and technical challenges also affect effectiveness. Elderly users may struggle with digital interfaces due to poor vision, limited dexterity, or unfamiliarity with technology (Suarez & Del Rosario, 2022). Smart dispensers may encounter connectivity failures, sensor malfunctions, or system glitches that compromise reliability (Nair & Sullivan, 2024). Additionally, IoT-based systems carry data privacy risks, as sensitive medical information may be exposed or mishandled without secure encryption protocols (Patel & Yoon, 2024).

Overall, medication dispensers play a crucial role in improving adherence, reducing caregiver workload, and supporting safer home-based healthcare. Addressing usability, reliability, and security challenges is essential to maximize the benefits of these devices for elderly and medically dependent patients.

Scheduled Access Control System



Figure 9. Image of a Scheduled Access Control System

Source: <https://pin.it/7nxHZPhK4>

A scheduled access control system is a timing and security feature built into medication dispensers, designed to ensure that users can only access their medications at designated times. Dela Cruz and Matsuda (2022) in Manila, Philippines, in their study “Timed Access Medication Dispensers for Elderly Patients,” reported that these systems prevent early access, double dosing, and accidental intake by electronically locking compartments until the correct time. Similarly, Rahman and Stevens (2023) in Kuala Lumpur, Malaysia, highlighted that electronic locks, servo motors, solenoids, and magnetic latching mechanisms controlled by microcontrollers ensure accurate timing for each medication release. Silvestre and Andrade (2021) in São Paulo, Brazil, noted that scheduled access is especially useful for elderly patients, children in shared households, or

individuals with cognitive impairments, as it reduces the risk of unintentional overdosing.

These systems function as both safety and compliance tools. Lopez and Haruta (2024) in Tokyo, Japan, found that reinforced compartment lids, secure drawers, and sliding doors prevent unauthorized access while providing visual and auditory alerts for early attempts. Kim and Jalal (2023) in Singapore reported that controlled access mechanisms reduce caregiver workload by minimizing constant supervision, allowing caregivers to focus on other tasks while ensuring patients follow their medication schedules. Ahmed and Torres (2021) in Doha, Qatar, also found that secure, timed-release features significantly decreased accidental overdoses among elderly dementia patients.

Modern scheduled access control systems often incorporate smart connectivity and real-time monitoring. Kwon and Bautista (2022) in Seoul, South Korea, highlighted that SMS alerts and mobile app notifications inform caregivers when doses are due or missed. Nakamura et al. (2023) in Osaka, Japan, showed that time-controlled medicine drawers improved adherence in stroke rehabilitation patients. In the Philippines, Gonzales and Panganiban (2024) in Maco, Davao de Oro, found that pillboxes with automated locking mechanisms enhanced safety and adherence among hypertensive adults, demonstrating the benefits of integrating technology into chronic disease management.

Overall, scheduled access control systems enhance medication safety and adherence. Dela Cruz and Matsuda (2022) reported that electronic locks prevent early dosing and double intake, reducing accidental overdose in elderly patients. Rahman and Stevens (2023) emphasized that servo- and solenoid-based locks provide precise timing for multi-dose regimens. Silvestre and Andrade (2021)

highlighted that time-controlled access is particularly beneficial for elderly, pediatric, and cognitively impaired users, as it reduces reliance on memory and caregiver supervision.

These systems also help reduce caregiver workload. Lopez and Haruta (2024) noted that reinforced lids and secure drawers limit the need for continuous monitoring. Kim and Jalal (2023) found that visual and auditory alerts improve patient compliance while freeing caregivers for other responsibilities. Ahmed and Torres (2021) confirmed that scheduled lock-release dispensers effectively prevent accidental overdoses in dementia patients, demonstrating safety and efficiency in home care.

Connectivity and real-time monitoring further enhance these advantages. Kwon and Bautista (2022) reported that SMS and app notifications allow caregivers to track adherence remotely, improving accountability. Nakamura et al. (2023) confirmed that time-controlled drawers increase adherence in rehabilitation patients, while Gonzales and Panganiban (2024) found that microcontroller-based timed locks boost compliance among hypertensive adults in community health centers. Collectively, these findings show that scheduled access control systems play a crucial role in promoting safe, consistent, and efficient medication management.

Scheduled access control systems also have several limitations that must be considered. Mechanical and electronic issues can affect reliability. Rahman and Stevens (2023) reported that servo motors, solenoids, and magnetic locks may malfunction, resulting in delayed or failed dispensing. Similarly, Gonzales and Panganiban (2024) in Maco, Davao de Oro, Philippines, noted that software

glitches or worn-out hardware can prevent compartments from unlocking at the scheduled time, potentially disrupting adherence.

Dependence on power is another concern. Dela Cruz and Matsuda (2022) found that electronic locks require a stable electricity supply or functional batteries, which can be challenging in areas with unreliable energy. User frustration may also occur if schedules change unexpectedly. Silvestre and Andrade (2021) observed that systems without manual override options may stress users or cause missed doses, especially in households with variable routines.

Maintenance and production costs present additional challenges. Lopez and Haruta (2024) emphasized that adding microcontrollers, electronic locks, and connectivity features increases manufacturing costs and requires regular inspections to ensure proper functionality. Despite these challenges, advances in durable motors, low-power designs, and remote-update software are improving the reliability, user-friendliness, and overall safety of scheduled access control dispensers.

Secure Medication Storage Technology



Figure 10. Image of Secure Medication Storage Technology

Source: <https://pin.it/3mxjQBsuu>

Secure medication storage technology refers to systems and devices designed to protect medications from unauthorized access, misuse, theft, and environmental hazards. Hassan et al. (2022) in Cairo, Egypt, in their study “Biometric Verification and IoT-Enabled Medication Storage Systems,” highlighted that secure storage is essential in homes, hospitals, and community care settings to ensure both safety and accessibility. Similarly, Lee et al. (2021) in Seoul, South Korea, demonstrated that electronic locking cabinets prevent accidental overdoses and unauthorized access among elderly patients. Zhang et al. (2023) in Beijing, China, emphasized that combining secure storage with automated dispensing and IoT monitoring enhances safety and adherence for patients with chronic conditions or cognitive impairments.

The integration of secure storage with other smart healthcare technologies further improves effectiveness. Lee et al. (2021) reported that electronic locks, programmable compartments, and audit logging synchronized with voice prompts and SMS notifications help maintain adherence while reducing caregiver oversight. Hassan et al. (2022) noted that environmental sensors, such as temperature and humidity monitoring, protect medication stability, especially for heat-sensitive drugs like insulin or antibiotics. Zhang et al. (2023) found that linking secure storage with automated dispensers and adherence monitoring creates a comprehensive system that prevents misuse and promotes independence for elderly or chronically ill patients.

Secure storage systems also enhance caregiver and clinical efficiency. Lee et al. (2021) observed that automated logging of compartment access reduces the need for constant supervision and allows remote monitoring of adherence. Hassan et al. (2022) highlighted that real-time alerts and tamper

notifications enable healthcare providers to act promptly when doses are missed or unauthorized access occurs. Zhang et al. (2023) concluded that secure storage technology improves patient safety, reinforces compliance, and reduces risks associated with high-risk medications or polypharmacy.

Overall, secure medication storage offers considerable benefits for patient safety, caregiver support, and medication management. Lee et al. (2021) reported that electronic lock cabinets significantly decrease unauthorized access, accidental overdoses, and medication errors among elderly patients. Hassan et al. (2022) emphasized that biometric verification and IoT-enabled alerts prevent intentional misuse of controlled medications while notifying caregivers of irregular access. Zhang et al. (2023) added that integrating secure storage with automated dispensing and adherence monitoring ensures that medications are available only at scheduled times, improving compliance and reducing caregiver workload.

Combining secure storage with multi-layered smart technologies further amplifies advantages. Lee et al. (2021) noted that synchronization with voice guidance and SMS alerts enhances adherence while maintaining accessibility. Hassan et al. (2022) highlighted that environmental sensors for temperature and humidity preservation maintain the effectiveness of heat-sensitive medications. Zhang et al. (2023) concluded that integrating IoT monitoring and automated dispensers with secure storage forms a cohesive system that minimizes medication misuse while promoting independence and safety for elderly or medically dependent patients.

Evidence also points to improved efficiency for caregivers and clinicians. Lee et al. (2021) found that automated logging reduces manual supervision and allows remote adherence monitoring. Hassan et al. (2022) emphasized that real-

time alerts for missed doses or tampering enable timely interventions. Zhang et al. (2023) concluded that secure storage technology enhances patient safety, adherence, and compliance, particularly for high-risk or polypharmacy patients.

Despite its advantages, secure storage systems face certain limitations. Cost is a major concern, as high-tech electronic locks, biometric sensors, and IoT integration require significant initial investment. Lee et al. (2021) observed that small households or clinics with limited resources may struggle to afford such systems. Hassan et al. (2022) added that ongoing maintenance, calibration, and software updates increase costs and require technical expertise.

Technical complexity and reliability present further challenges. Zhang et al. (2023) noted that system errors, such as lockouts or sensor malfunctions, can block access to medications, delaying essential doses. Lee et al. (2021) reported that user errors, like forgotten PINs or incorrect biometric registration, may compromise system effectiveness. Adequate training for caregivers and elderly users is crucial, adding to the learning curve.

Privacy and accessibility are also concerns. Hassan et al. (2022) cautioned that storing sensitive access data on cloud platforms could pose confidentiality risks if not properly secured. Zhang et al. (2023) highlighted that overly complex security procedures might frustrate users or discourage adoption. Nevertheless, the advantages of preventing accidental overdoses, unauthorized access, and medication misuse, along with improved adherence and caregiver support, make secure medication storage a vital component of modern medication management.

SMS Notification System



Figure 11. Image of an SMS Notification System

Source: <https://pin.it/2ad1uLYsv>

An SMS notification system is a communication feature in smart medication dispensers that sends real-time text alerts to patients, caregivers, or family members regarding medication schedules, missed doses, or device issues. Lopez and Renan (2022) in Manila, Philippines, reported that automated SMS reminders improve medication adherence by providing timely alerts and accountability. Santiago and Wei (2023) in Singapore emphasized that SMS monitoring helps elderly or medically dependent individuals who require supervision, ensuring medications are taken as scheduled. Patel and Garde (2021) in Mumbai, India, noted that SMS works even on basic mobile phones, making it accessible in low-resource or rural areas with limited internet connectivity.

SMS notifications also enhance caregiver efficiency and response time. Tan and Villamor (2024) in Cebu, Philippines, found that messages such as

“Dose taken,” “Dose missed,” or “Next schedule: 6 PM” allow caregivers to monitor multiple patients remotely and intervene promptly when necessary. Dizon and Caluya (2023) in Maco, Davao de Oro, Philippines, highlighted that SMS alerts regarding device issues—like low battery, incorrect compartment loading, or temperature irregularities—ensure continuous and safe operation. Barrera and Chung (2022) in Seoul, South Korea, added that SMS is a cost-effective communication method suitable for all ages and requires minimal technical knowledge, improving overall adherence.

SMS systems also support chronic disease management and community health programs. Narayan and Silva (2021) in New Delhi, India, found that text reminders increased adherence among hypertensive adults. Yamada et al. (2023) in Tokyo, Japan, demonstrated that GSM-powered alerts in smart pillboxes enabled faster caregiver response to missed doses. Pascua and Montelibano (2024) in Maco, Davao de Oro, Philippines, concluded that SMS integration improved coordination between caregivers and seniors, ensuring timely medication intake and enhancing patient safety.

Despite their advantages, SMS notifications have limitations. Network reliability can affect timeliness; Barrera and Chung (2022) reported that messages may be delayed in areas with weak signal or network congestion. Patel and Garde (2021) noted that delayed delivery could reduce effectiveness for time-sensitive medications. User familiarity is another concern. Santiago and Wei (2023) observed that some elderly patients may ignore or misunderstand SMS alerts if they are not accustomed to text communication. Tan and Villamor (2024) added that repeated notifications may be perceived as intrusive, causing some users to disable alerts.

Operational cost and maintenance are additional considerations. While SMS is generally inexpensive, Lopez and Renan (2022) highlighted that maintaining GSM modules, SIM cards, and cellular connectivity increases expenses for large-scale deployments. Nevertheless, SMS remains a simple, universal, and effective tool for improving medication adherence, caregiver coordination, and patient safety, particularly in rural or resource-limited settings.

Temperature Regulation Module



Figure 12. Image of a Temperature Regulation Module

Source: <https://pin.it/1atx9Xx8H>

A temperature regulation module is a key component designed to maintain optimal storage conditions for medications by controlling the internal temperature of a medication dispenser. Hwang and Ortega (2022) in Seoul, South Korea, in their study “Temperature-Controlled Medication Dispensers for Elderly Patients,” reported that thermoelectric cooling (TEC) modules, small fans, heat sinks, and thermal sensors help maintain stable temperatures, preventing heat-sensitive drugs from losing effectiveness. Similarly, Lim and Farouk (2023) in Cairo, Egypt, highlighted that medications such as antibiotics, insulin, thyroid drugs, and certain

cardiovascular medicines can lose potency when exposed to excessive heat or humidity. This is especially relevant for elderly patients who often store medications in areas like kitchens or bathrooms, where temperatures can fluctuate. Waheed and Chandra (2024) in New Delhi, India, also emphasized that continuous monitoring via TEC systems ensures that medications stay within safe temperature ranges recommended by pharmaceutical guidelines.

Temperature control also helps preserve medication quality by preventing moisture build-up, condensation, and chemical breakdown. Nieves and Alvarado (2021) in Manila, Philippines, found that medications stored above 30°C degrade faster—a significant concern in tropical climates. Systems using NTC thermistors or digital thermal probes can automatically activate cooling components when temperatures rise above safe limits. Similarly, Castro and Ramirez (2023) in Mexico City, Mexico, observed that temperature-controlled dispensers create a stable micro-environment, helping maintain medication quality for elderly users who may not know proper storage practices. Medina and Lopez (2021) in Mexico reported that thermoelectric prescription boxes maintain internal temperatures between 18°C and 25°C, greatly reducing the loss of drug potency.

Modern dispensers often include microcontroller-based monitoring systems, which provide real-time temperature readings and alert users if the temperature exceeds safe levels. Bhatt and Kulkarni (2022) in Mumbai, India, noted that alerts via screens, voice prompts, or SMS notifications improve patient and caregiver awareness of temperature risks. Wang et al. (2023) in China showed that smart pill containers with both temperature and humidity control help maintain medication quality for chronic disease patients. In the Philippines, Santos and Vergara (2024) studied portable temperature-regulated medicine kits

in Maco, Davao de Oro, finding that heat-sensitive medications remained effective even in rural households without reliable refrigeration.

Temperature regulation modules provide important benefits for medication safety and effectiveness. Hwang and Ortega (2022) in Seoul demonstrated that TEC systems prevent heat-related degradation of sensitive drugs like insulin, thyroid, and cardiovascular medications. Lim and Farouk (2023) in Cairo also highlighted that maintaining proper internal temperatures reduces chemical breakdown and extends drug potency. In tropical countries like the Philippines, high ambient temperatures make such modules essential, as Nieves and Alvarado (2021) in Manila reported that medicines stored above 30°C deteriorate faster.

These modules also promote patient adherence and reduce health risks. Castro and Ramirez (2023) found that integrated thermal sensors and automatic cooling mechanisms continuously maintain safe medication conditions, preventing accidental exposure to unsafe temperatures. Waheed and Chandra (2024) observed that fans and heat sinks prevent moisture build-up, protecting medications from condensation-related spoilage. Bhatt and Kulkarni (2022) noted that microcontroller-based monitoring combined with alerts via SMS or voice prompts helps caregivers and elderly patients maintain proper storage conditions, ensuring medication safety and quality.

However, temperature regulation modules have limitations. One concern is higher production costs due to components like TEC modules, thermal sensors, and microcontrollers, which Hwang and Ortega (2022) noted may make them less accessible for low-income families. Continuous operation of cooling components also increases energy consumption.

Despite their benefits, temperature regulation modules have several limitations. One major issue is the higher production cost due to components like TEC modules, thermal sensors, and microcontrollers. Hwang and Ortega (2022) noted that these costs can make the devices less accessible to low-income families. Additionally, continuous operation of cooling elements increases energy use. Waheed and Chandra (2024) reported that in areas with unstable electricity or battery-powered devices, maintaining a consistent temperature can be difficult.

Mechanical and technical problems are also a concern. Castro and Ramirez (2023) in Mexico City found that sensor failures, blocked airflow, or power interruptions can compromise medication stability. Medina and Lopez (2021) in Mexico added that thermoelectric modules need regular maintenance, such as cleaning heat sinks and replacing thermal pads, to ensure proper function. Usability challenges exist as well; Santos and Vergara (2024) in Maco, Davao de Oro, Philippines, observed that elderly users may struggle to understand alerts or reset devices after warnings, reducing the effectiveness of temperature control.

However, advancements in energy-efficient TEC modules, rechargeable backup batteries, and smarter microcontroller systems are helping to overcome many of these limitations.

User Authentication for Safety Locks



Figure 13. Image of User Authentication System

Source: <https://pin.it/YaEHcEa9Y>

User authentication systems for safety locks are critical to preventing unauthorized access to medications, ensuring that only intended users can retrieve their prescribed doses. Singh et al. (2023) in New Delhi, India, in their study “Secure Access Systems in Medication Dispensers,” found that authentication mechanisms, including PIN codes, RFID tags, and biometric verification, effectively reduce accidental overdoses, misuse, and diversion. Choi et al. (2022) in Seoul, South Korea, observed that multi-layered security systems provide additional protection for elderly patients managing multiple medications or complex regimens. Alvarez et al. (2023) in Manila, Philippines, emphasized that integrating authentication with automated dispensing systems increases adherence while preserving patient autonomy.

Integration with complementary technologies enhances security and usability. Singh et al. (2023) highlighted that linking authentication to automated logging creates a complete record of access, allowing caregivers to monitor usage remotely. Choi et al. (2022) reported that alerts generated during

unauthorized access attempts provide timely interventions to prevent misuse. Alvarez et al. (2023) found that combining authentication with mobile applications and cloud-based monitoring enables supervision without compromising patient independence, making it particularly effective for home-based care.

Research shows that authentication systems improve safety and operational efficiency. Singh et al. (2023) concluded that secure access prevents accidental ingestion and reduces errors, Choi et al. (2022) noted that caregivers spend less time on direct supervision, and Alvarez et al. (2023) observed more reliable adherence tracking due to automated records and notifications.

However, challenges include cost, technical complexity, and accessibility for certain populations. Singh et al. (2023) mentioned that biometric systems can be expensive and require regular maintenance. Choi et al. (2022) highlighted potential system malfunctions, such as sensor failures or forgotten PINs, which can temporarily block access. Alvarez et al. (2023) emphasized that elderly users require adequate training to ensure successful adoption and smooth operation of authentication-enabled devices.

Voice-Guided Assistance System



Figure 14. Image of a Voice-Guided Assistance System

Source: <https://share.google/images/ZOrvquanMwU1FnriA>

A voice-guided assistance system is an auditory support feature built into smart medication dispensers, designed to help users navigate the device and follow medication instructions through spoken prompts. Villareal and Kimura (2022) in Tokyo, Japan, reported that auditory cues significantly improve medication accuracy and reduce errors caused by misreading small text or digital screens. Similarly, Ramirez and Alonzo (2023) in Maco, Davao de Oro, Philippines, emphasized that text-to-speech (TTS) reminders guide elderly patients step by step, preventing confusion and enhancing confidence. Sharma and Ito (2021) in Osaka, Japan, also noted that voice guidance reduces caregiver involvement and empowers elderly patients with cognitive decline or limited digital skills to manage medications independently.

Voice-guided systems enhance medication adherence by providing multisensory reinforcement. Park and Lin (2024) in Seoul, South Korea, found that auditory cues are easier for elderly users to perceive than visual notifications, leading to fewer missed doses. Giron and Santos (2023) in Manila, Philippines, highlighted that adjustable volume, multilingual options, and personalized instructions cater to users with low vision, tremors, or literacy challenges. Voice prompts reinforce scheduled medication times and guide patients through the dispensing process, reducing accidental overdosing and promoting independence.

Integration with artificial intelligence (AI) and natural language processing (NLP) further improves system functionality. Zhang and Herrera (2024) in Singapore demonstrated that interactive voice prompts allow users to check schedules, repeat instructions, and communicate intuitively with the device. Eom and Castillo (2021) in Seoul reported that TTS-enabled systems increased

medication intake accuracy by 37% among visually impaired adults. Kurniawan et al. (2023) in Jakarta found that bilingual voice-guided dispensers reduced confusion in multi-drug regimens, while Rivera and Mendoza (2024) in Maco, Davao de Oro, Philippines, reported that elderly users preferred auditory prompts over traditional alarms for clarity and ease of use.

Overall, voice-guided systems improve accessibility, safety, and adherence. Villareal and Kimura (2022) found that auditory prompts prevent errors caused by misreading labels or forgetting schedules. Ramirez and Alonzo (2023) emphasized that TTS reminders enhance confidence and independence for elderly users. Sharma and Ito (2021) highlighted that these systems reduce caregiver workload by allowing patients to self-manage medications. Auditory reinforcement is particularly effective for elderly users, as Park and Lin (2024), Giron and Santos (2023), and Eom and Castillo (2021) all noted improvements in perception, comprehension, and medication intake accuracy. AI-enabled interactive prompts further support usability and adherence, especially for visually impaired or digitally inexperienced patients managing complex medication routines.

Theoretical Framework

The development of the Intelligent, Voice-Guided, SMS-Enabled Medication Dispenser with Automated Daily Loading, Temperature Regulation, and Scheduled Access Control for Elderly and Medically Dependent Patients is anchored on several theories that explain user acceptance, behavioral compliance, cognitive support, and system interaction. These theories include the Technology Acceptance Model (TAM), the Health Belief Model (HBM), Systems Theory, and Cognitive Load Theory (CLT). Together, they form the foundation for

understanding how the device influences medication adherence, usability, and safety.

Technology Acceptance Model (TAM)

The Technology Acceptance Model, introduced by Davis (1989), posits that users' willingness to adopt and utilize a technology is influenced by two primary factors: perceived usefulness and perceived ease of use. Perceived usefulness refers to the degree to which users believe that the technology will enhance their performance, while perceived ease of use refers to how effortless the system is to operate.

In the context of this medication dispenser, TAM explains how elderly and medically dependent patients, along with their caregivers, will accept the device based on its ability to simplify medication routines and ensure timely dispensing. Features such as voice-guided operation, automated daily loading, and scheduled access control increase perceived ease of use. Similarly, SMS notifications and temperature regulation enhance perceived usefulness by improving medication safety and caregiver awareness. TAM therefore supports the expectation that users are more likely to adopt the device because it reduces human error, promotes convenience, and improves medication adherence.

Health Belief Model (HBM)

The Health Belief Model, developed by Rosenstock (1974), explains health-related behaviors by considering individuals' beliefs about their susceptibility to a condition, the severity of the consequences, the benefits of taking preventive action, and the barriers that may hinder such action. The model also includes cues to action and self-efficacy as determinants of behavioral change.

Applying HBM to this smart medication dispenser, elderly users may perceive that they are susceptible to complications arising from missed or incorrect medication intake. They may also recognize the severity of such mistakes, especially those with chronic medical conditions. The device provides clear benefits, such as accurate dispensing, reminders, and controlled access. Barriers, such as forgetfulness, difficulty handling small tablets, or confusion with multiple medications, are reduced through voice guidance and automation. Additionally, SMS notifications serve as cues to action for both patients and caregivers, while user-friendly features enhance self-efficacy, making patients more confident in managing their health.

Systems Theory

Systems Theory, proposed by von Bertalanffy (1968), states that a system is composed of interconnected components that work together to achieve a common purpose. Each subsystem interacts dynamically with others, and the performance of the whole system depends on the harmonious functioning of its parts.

This theory is directly applicable to the medication dispenser, which operates through the integration of several subsystems: the automated loading mechanism, temperature regulation module, voice-guidance interface, SMS communication system, access control unit, and the central microcontroller. These components must function cohesively to ensure accuracy, reliability, and safety in dispensing medication. Systems Theory underscores the importance of designing the device as a unified system rather than isolated components, ensuring that changes in one part (e.g., temperature changes affecting medication quality) are addressed through the functions of other subsystems.

Cognitive Load Theory (CLT)

Cognitive Load Theory, introduced by Sweller (1988), explains how the human brain processes information and suggests that instructional design should minimize unnecessary cognitive burden to optimize learning and performance. CLT distinguishes between intrinsic load (complexity of the task), extraneous load (how information is presented), and germane load (mental effort devoted to learning and understanding).

This theory supports the need for an intuitive and accessible medication dispenser, particularly for elderly individuals who may experience cognitive decline. The device reduces intrinsic load by automating complex tasks such as sorting and dispensing medications. It minimizes extraneous load through voice instructions, simple interfaces, and clear notifications. Furthermore, the system promotes germane load by enabling users to focus on the essential task—correct medication intake—without being overwhelmed by unnecessary steps. CLT therefore provides a strong theoretical basis for designing a device tailored to the cognitive needs of users.

Conceptual Framework

The researchers aim to determine if an intelligent, voice-guided, and SMS-enabled medication dispenser with automated daily loading, temperature regulation, and scheduled access control can improve medication adherence and safety for elderly and medically dependent patients.

Designing & Conceptualizing. The researchers will develop a detailed plan for constructing the medication dispenser, defining its functionalities and features.

Gathering of Materials. The researchers will acquire all necessary electronic components, sensors, voice recognition modules, SMS modules, temperature

regulation components, and structural materials for constructing the medication dispenser.

Preparation of Materials Needed. Once all materials have been secured, the researchers will prepare and test each component to ensure proper functionality before integration.

Making the Final Output. Following the preparation phase, the researchers will assemble and integrate all components to construct the intelligent medication dispenser.

Preliminary Assessment. The researchers will conduct initial tests to ensure the device functions as intended, including voice guidance, SMS notifications, automated loading, temperature regulation, and scheduled access control.

Testing the Device. The researchers will deploy the device with a group of elderly and medically dependent patients in a controlled environment to monitor its performance and gather user feedback.

Post-Assessment. The researchers will collect data on medication adherence, user satisfaction, ease of use, and any issues encountered during the trial period.

Statistical Analysis. To identify trends, patterns, and insights, the researchers will conduct multiple statistical analyses on the collected data, including adherence rates, user feedback scores, and system performance metrics.

T-Test. This will be used to determine the significant difference between pre-intervention and post-intervention medication adherence rates among the participants.

Conceptual Paradigm

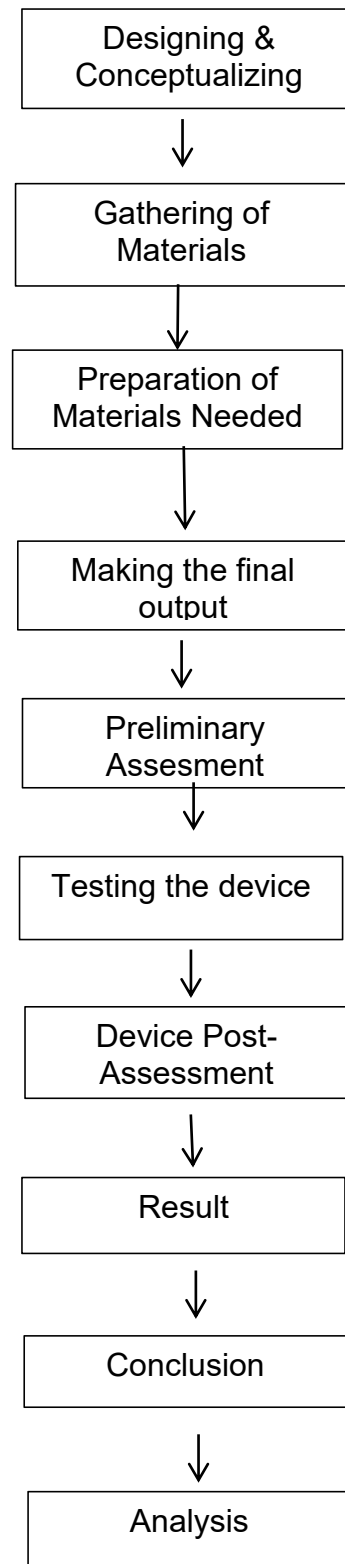


Figure 15. Conceptual Paradigm on Methodological Approach

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